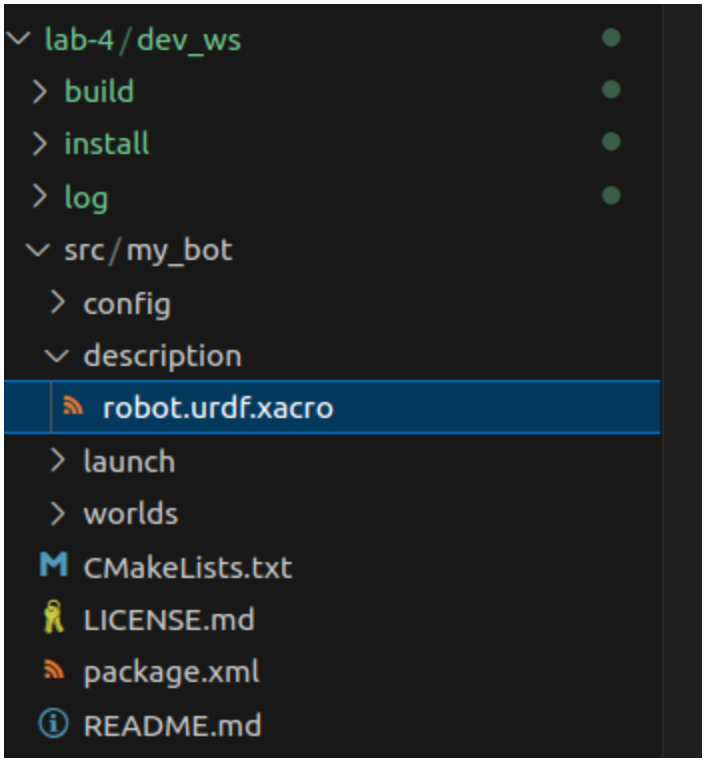
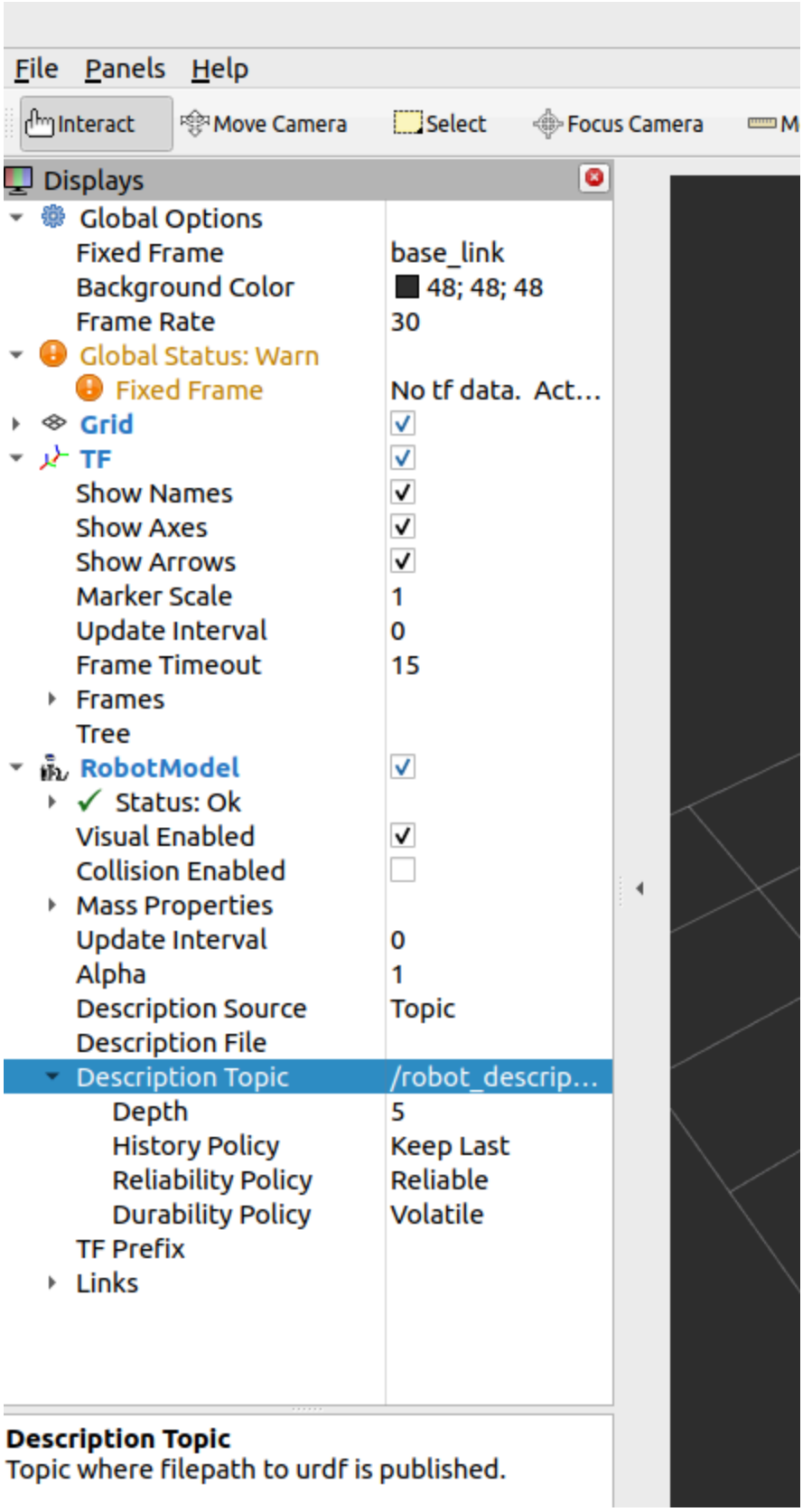
MAR LAB 4

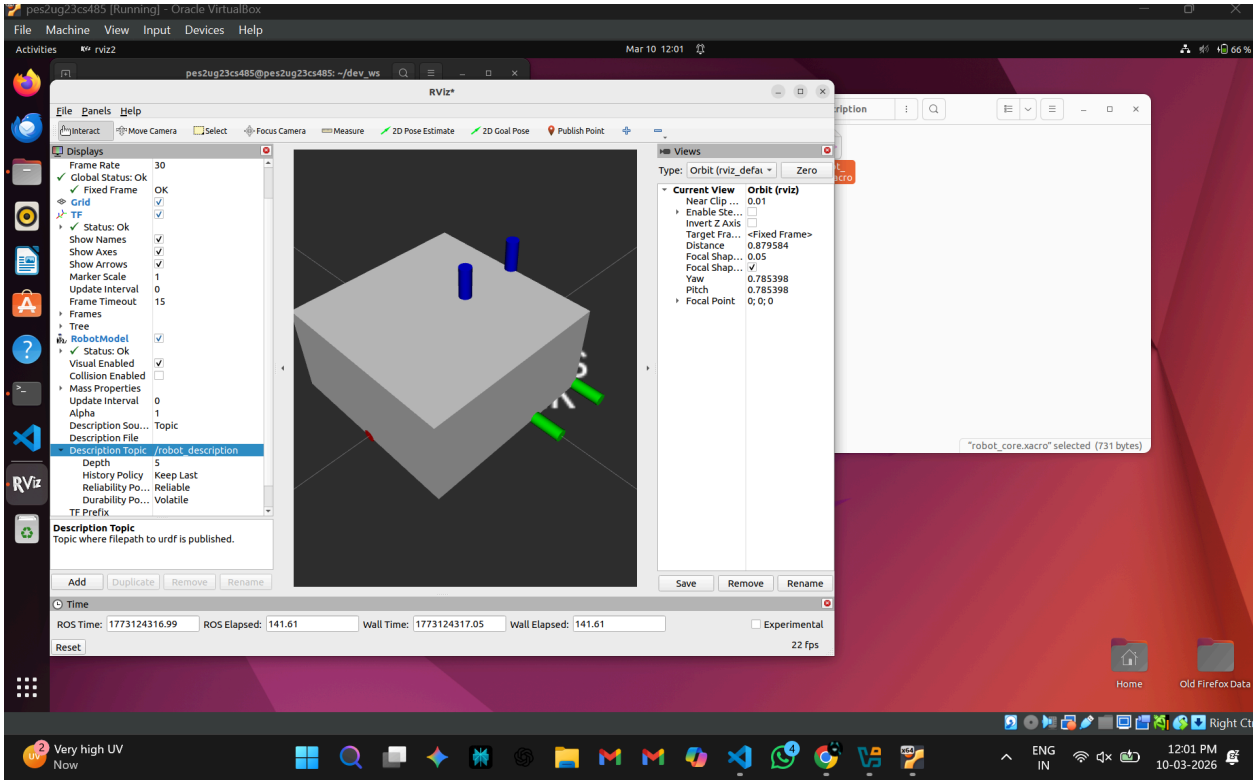
Name: Rithvik Rajesh Matta
Class: 6H
SRN: PES2UG23CS485

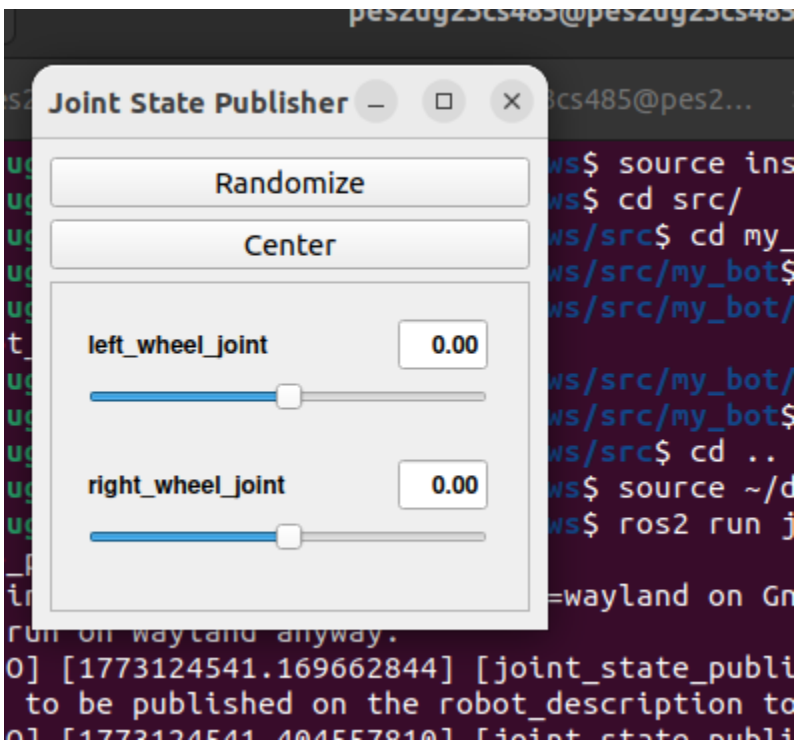
```
pes2ug23cs485@pes2ug23cs485:~/sem6/MAR/lab-4$ mkdir dev_ws
cd dev_ws
mkdir src
cd src
git clone https://github.com/joshnewans/my_bot.git
cd ..
colcon build --symlink-install
Cloning into 'my_bot'...
remote: Enumerating objects: 20, done.
remote: Counting objects: 100% (10/10), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 20 (delta 3), reused 1 (delta 1), pack-reused 10 (from 1)
Receiving objects: 100% (20/20), 8.08 KiB | 8.08 MiB/s, done.
Resolving deltas: 100% (3/3), done.
Starting >>> my_bot
Finished <<< my_bot [1.56s]

Summary: 1 package finished [2.21s]
```

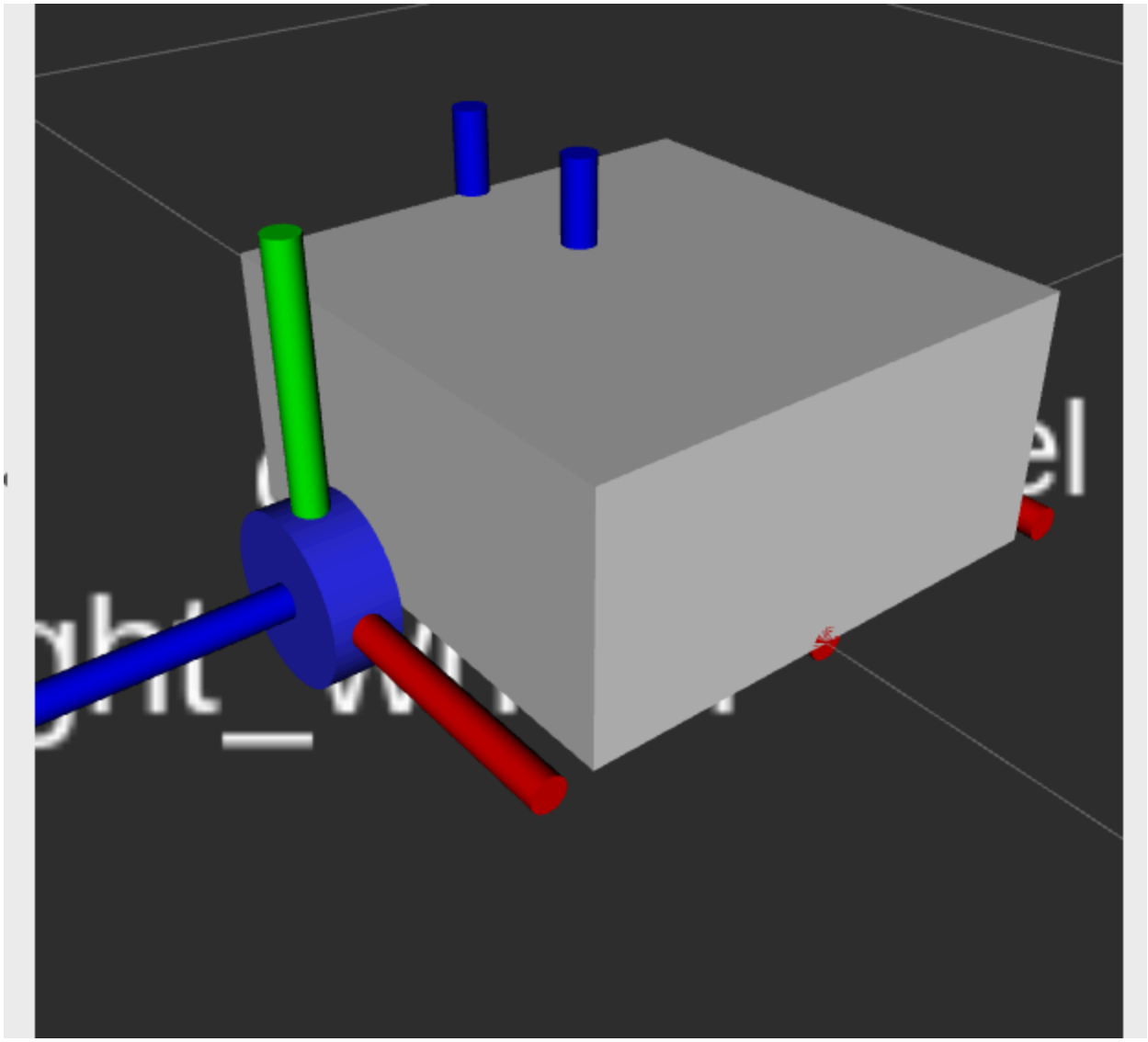
```
pes2ug23cs485@pes2ug23cs485:~/sem6/MAR/lab-4/dev_ws$ source install/setup.bash
pes2ug23cs485@pes2ug23cs485:~/sem6/MAR/lab-4/dev_ws$ ros2 launch my_bot rsp.laun
ch.py
[INFO] [launch]: All log files can be found below /home/pes2ug23cs485/.ros/log/2
026-03-09-11-38-59-403325-pes2ug23cs485-4320
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [robot_state_publisher-1]: process started with pid [4321]
[robot_state_publisher-1] [INFO] [1773036539.736378616] [robot_state_publisher]:
 got segment base_link
```

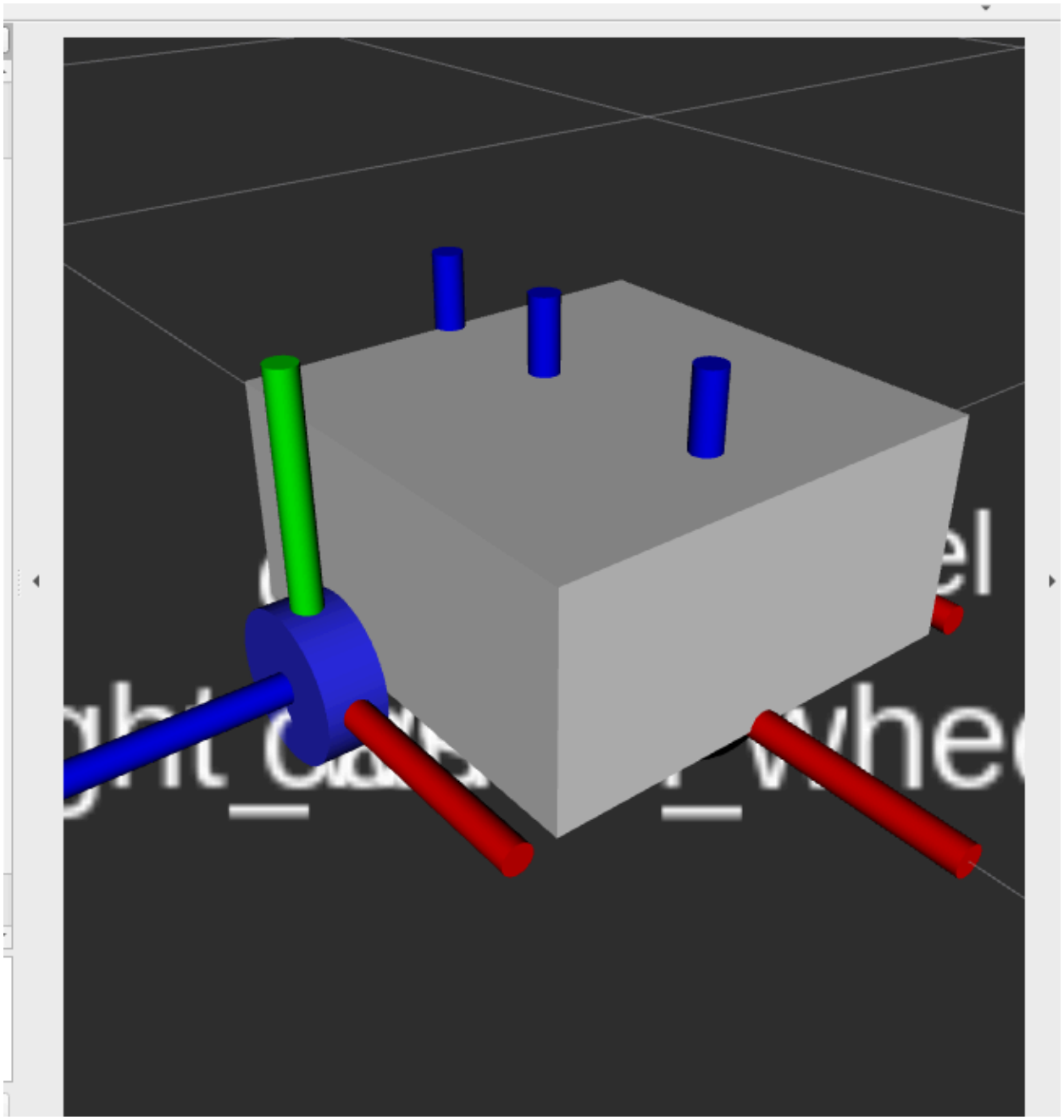

lab-4 / dev_ws
build
install
log
src / my_bot
config
description
robot.urdf.xacro
launch
worlds
CMakeLists.txt
LICENSE.md
package.xml
README.md

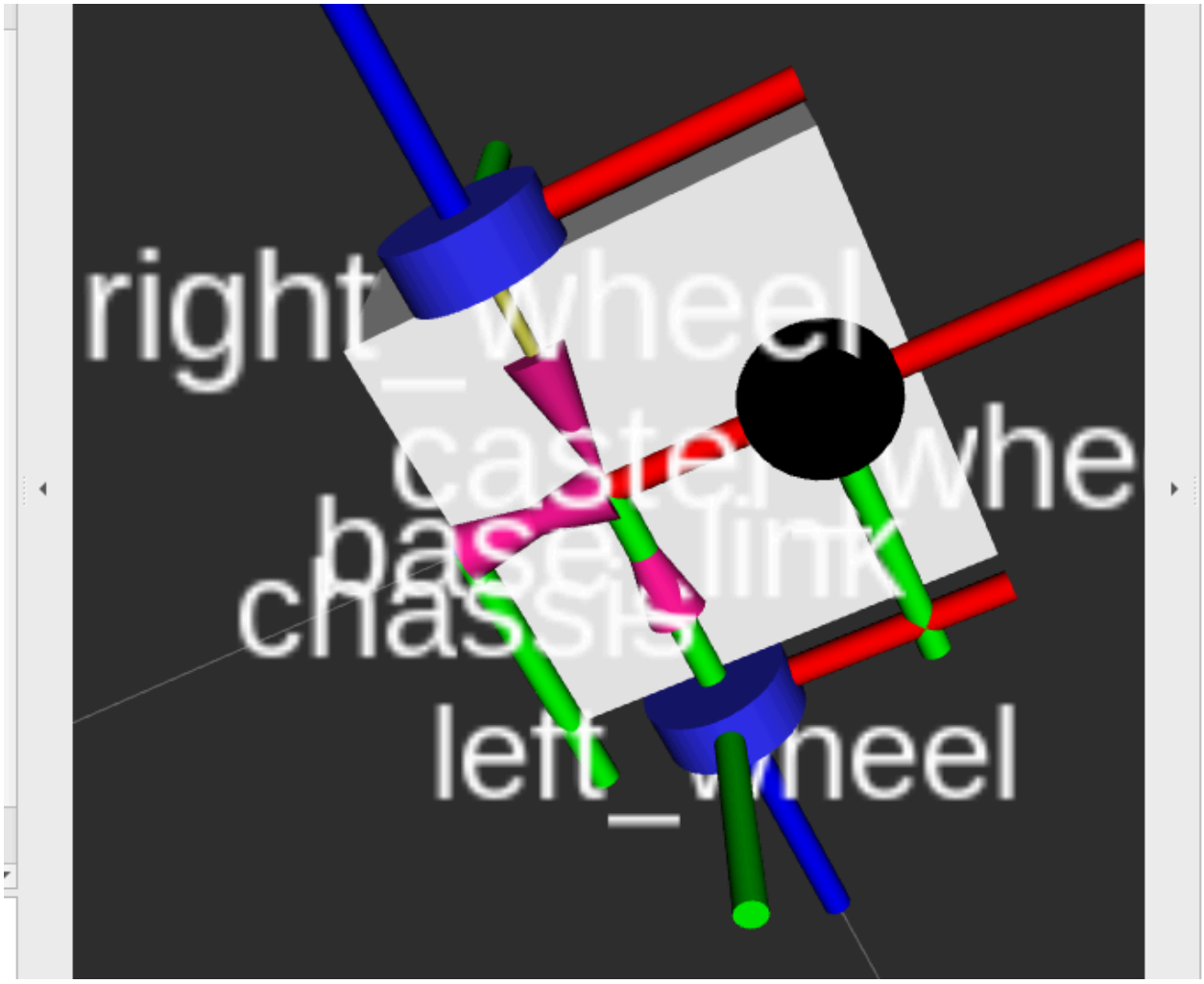

File
Panels
Help
Interact
Move Camera
Select
Focus Camera
Displays
Global Options
Fixed Frame
base_link
Background Color
48; 48; 48
Frame Rate
30
Global Status: Warn
Fixed Frame
No tf data. Act...
Grid
TF
Show Names
Show Axes
Show Arrows
Marker Scale
1
Update Interval
0
Frame Timeout
15
Frames
Tree
RobotModel
Status: Ok
Visual Enabled
Collision Enabled
Mass Properties
Update Interval
0
Alpha
1
Description Source
Topic
Description File
Description Topic
/robot_descrip...
Depth
5
History Policy
Keep Last
Reliability Policy
Reliable
Durability Policy
Volatile
TF Prefix
Links
Description Topic
Topic where filepath to urdf is published.

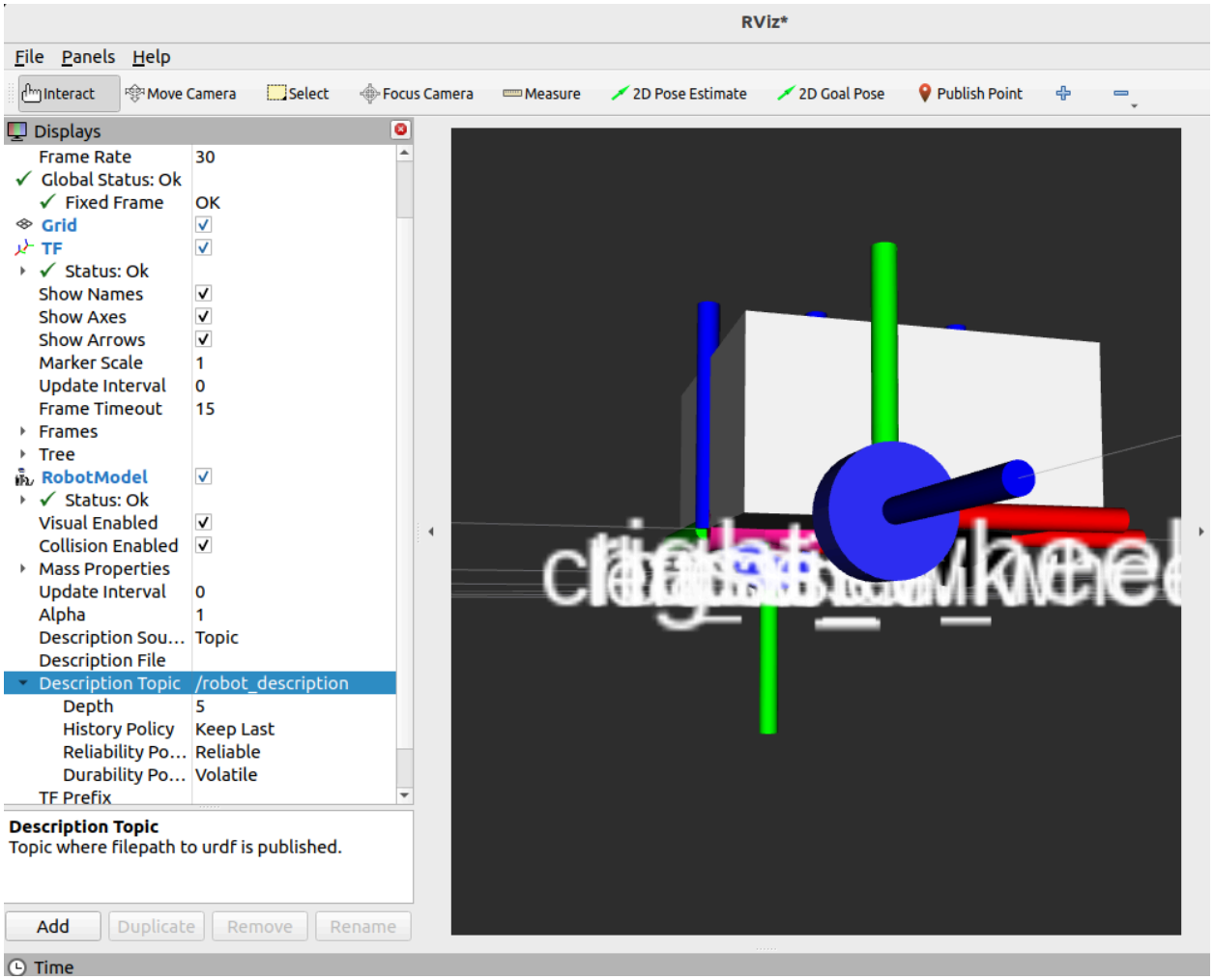

pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities
rviz2
Mar 10 12:01
pes2ug23cs485@pes2ug23cs485: ~/dev_ws
RViz*
File Panels Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
Displays
Frame Rate 30
Global Status: Ok
Fixed Frame OK
Grid
TF
Status: Ok
Show Names
Show Axes
Show Arrows
Marker Scale 1
Update Interval 0
Frame Timeout 15
Frames
Tree
RobotModel
Status: Ok
Visual Enabled
Collision Enabled
Mass Properties
Update Interval 0
Alpha 1
Description Sou... Topic
Description File
Description Topic /robot_description
Depth 5
History Policy Keep Last
Reliability Po... Reliable
Durability Po... Volatile
TF Prefix
Description Topic
Topic where filepath to urdf is published.
Add
Duplicate
Remove
Rename
Views
Type: Orbit (rviz_defa
Zero
Current View Orbit (rviz)
Near Clip ... 0.01
Enable Ste...
Invert Z Axis
Target Fra... <Fixed Frame>
Distance 0.879584
Focal Shap... 0.05
Focal Shap...
Yaw 0.785398
Pitch 0.785398
Focal Point 0; 0; 0
Save
Remove
Rename
Time
ROS Time: 1773124316.99
ROS Elapsed: 141.61
Wall Time: 1773124317.05
Wall Elapsed: 141.61
Experimental
Reset
22 fps
"robot_core.xacro" selected (731 bytes)
Home
Old Firefox Data
Right Ct
Very high UV
Now
ENG
IN
12:01 PM
10-03-2026


Joint State Publisher
Randomize
Center
left_wheel_joint
0.00
right_wheel_joint
0.00
ws$ source ins
ws$ cd src/
ws/src$ cd my_
ws/src/my_bot$
ws/src/my_bot/
ws/src/my_bot/
ws/src/my_bot$
ws/src$ cd ..
ws$ source ~/d
ws$ ros2 run j
=wayland on Gn
run on wayland anyway.
0] [1773124541.169662844] [joint_state_publi
to be published on the robot_description to

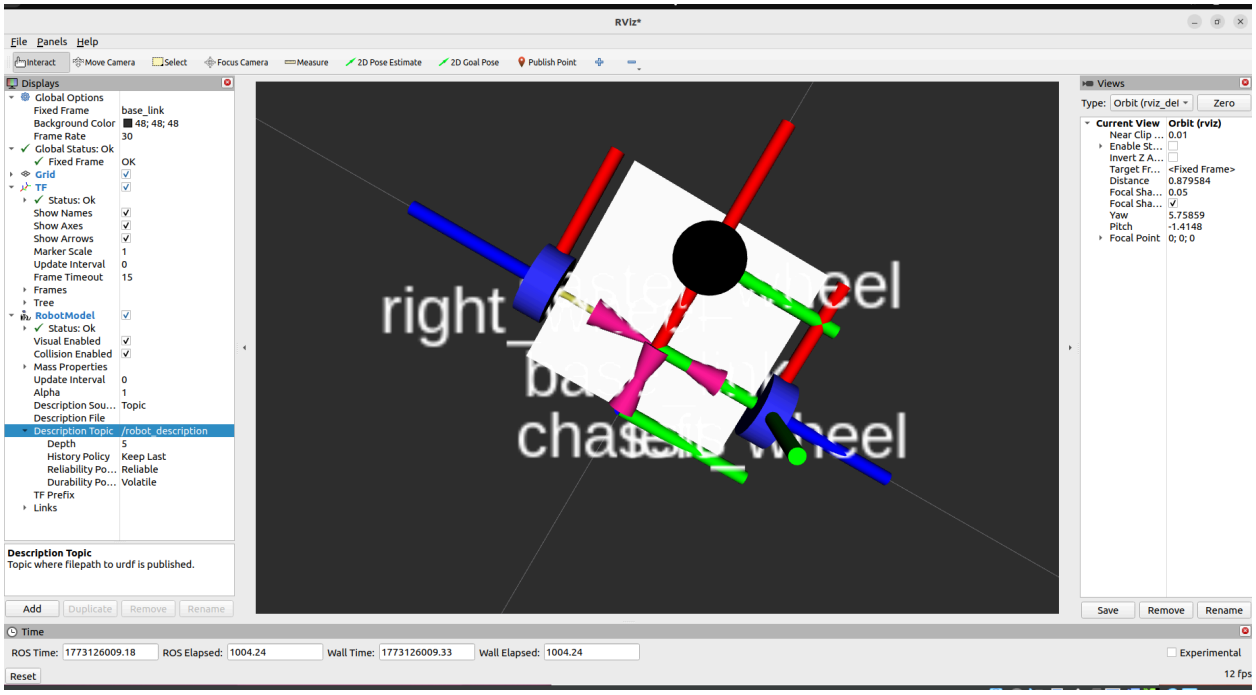




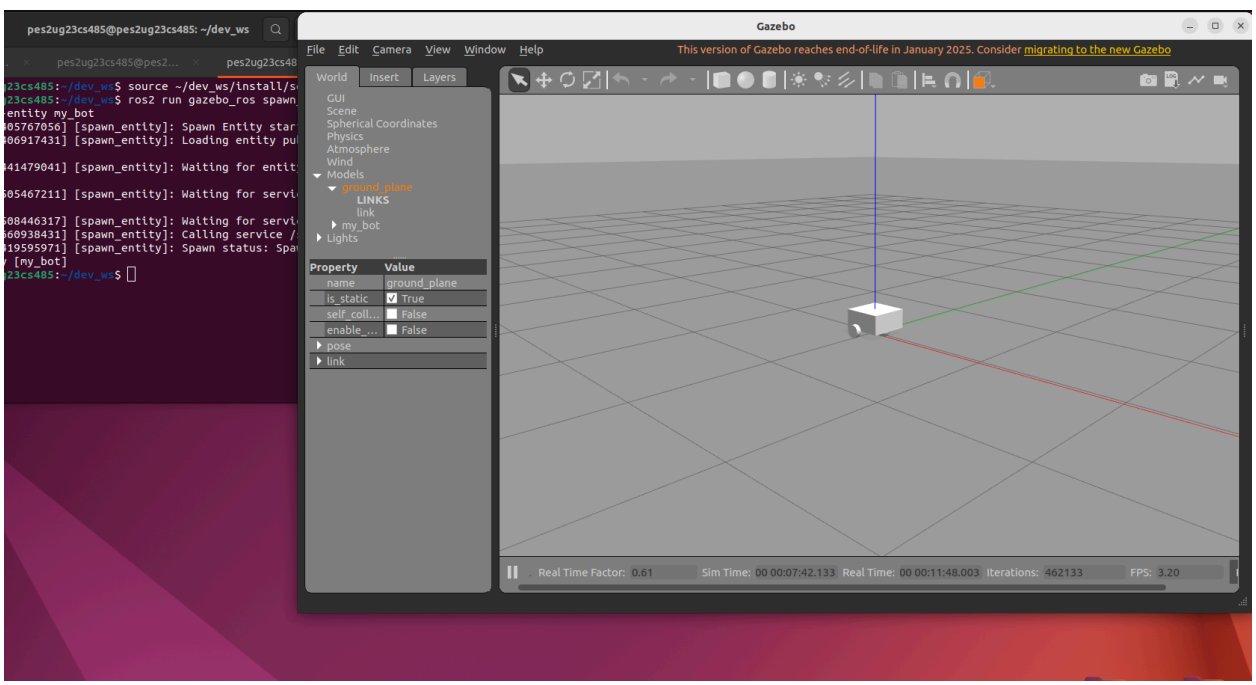

right_wheel
caster_whe
base_link
chassis
left_wheel

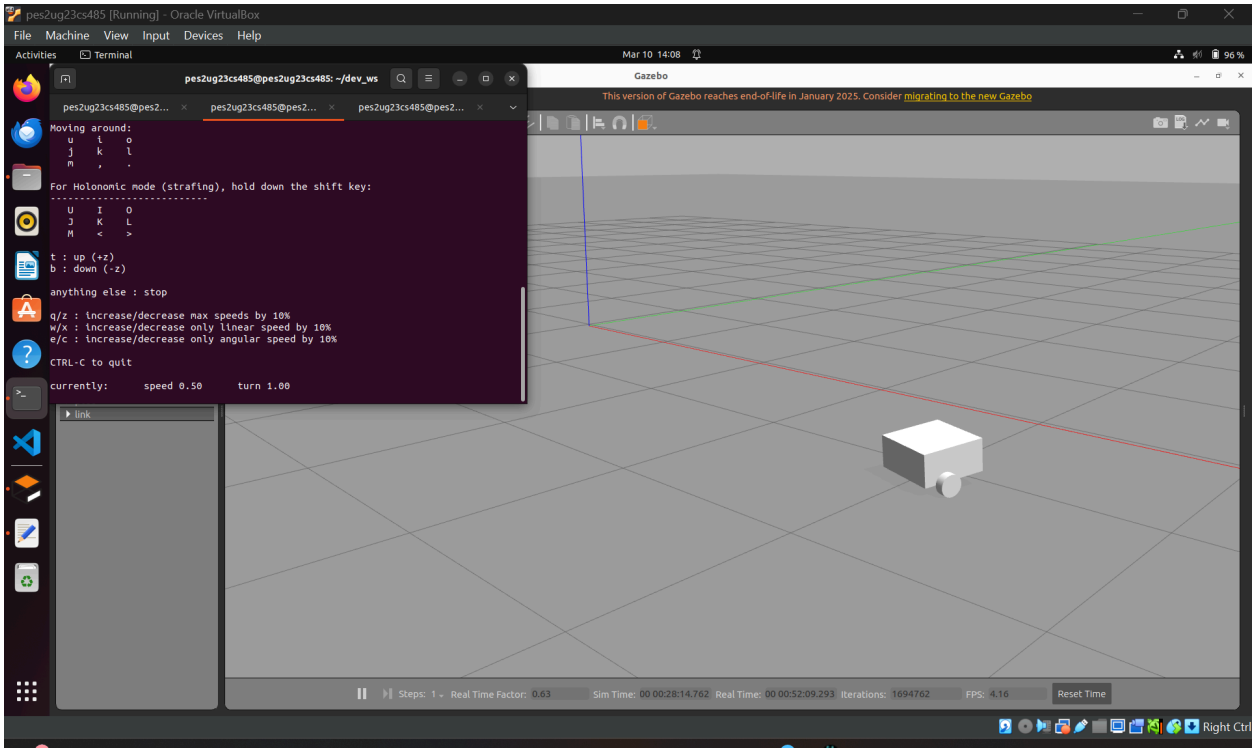


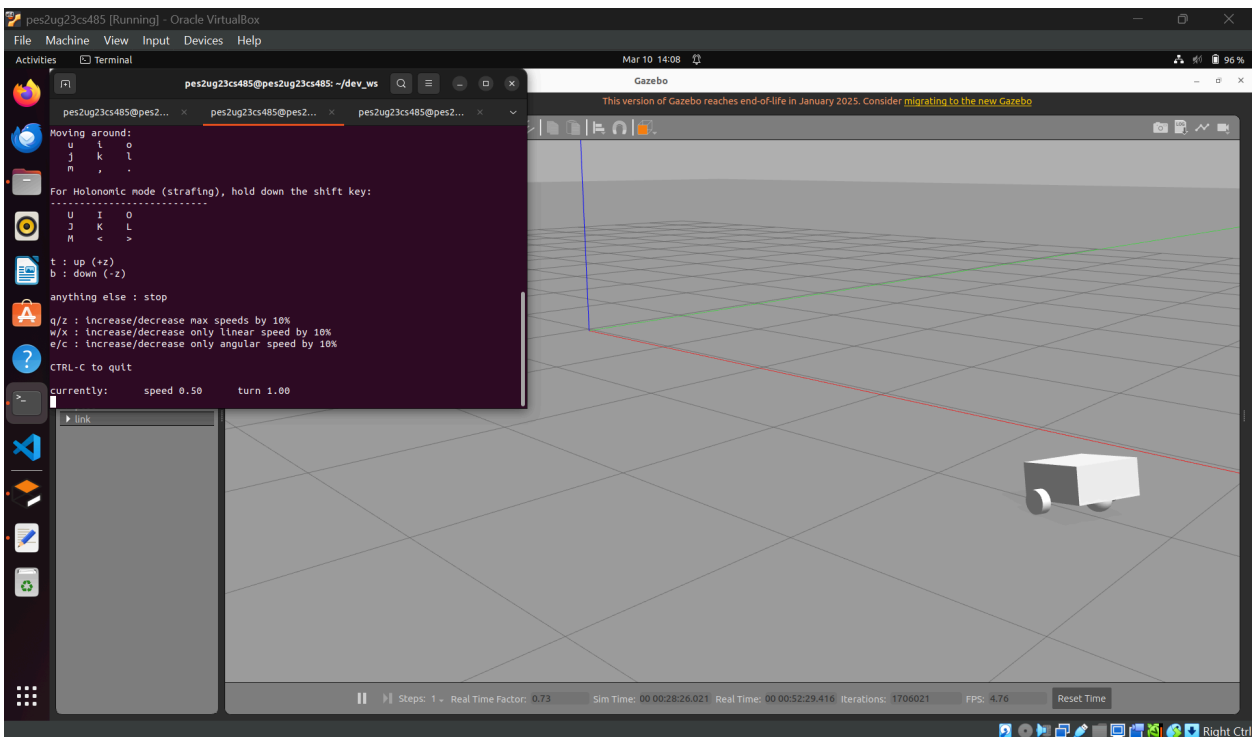
Collision enabled

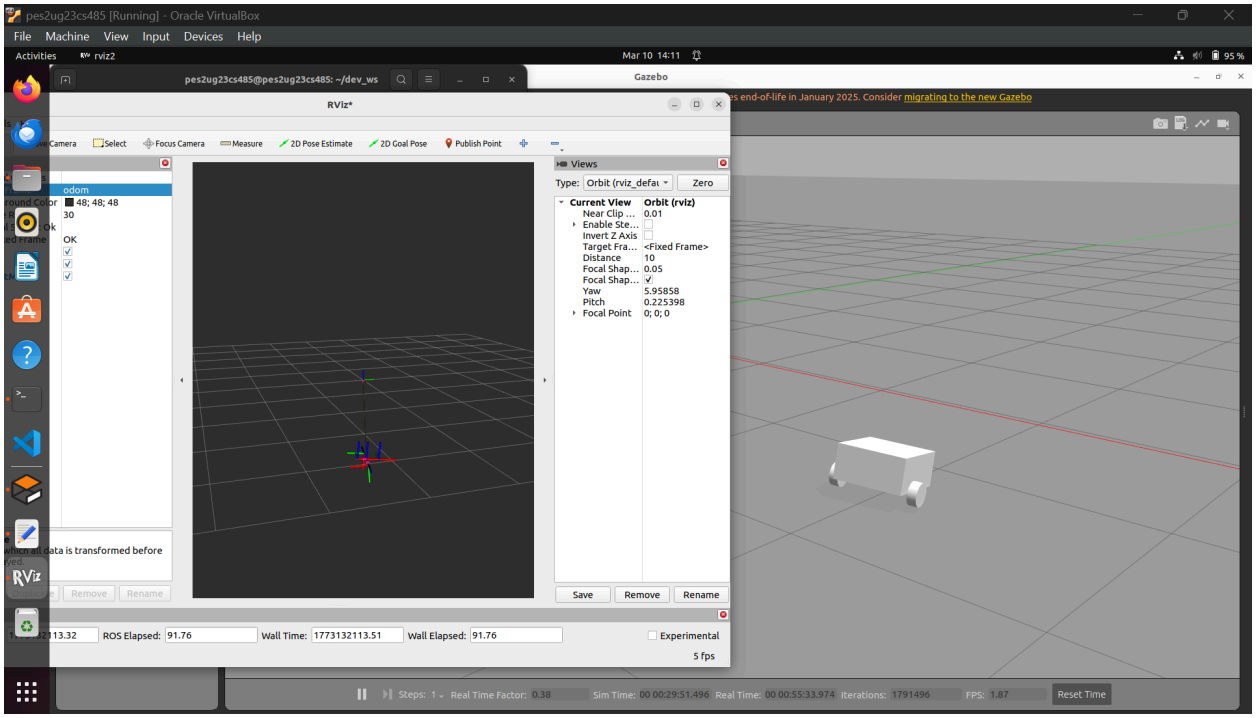

RVIz*
File Panels Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
Displays
Global Options
Fixed Frame base_link
Background Color 48; 48; 48
Frame Rate 30
Global Status: Ok
Fixed Frame OK
Grid
TF
Status: Ok
Show Names
Show Axes
Show Arrows
Marker Scale 1
Update Interval 0
Frame Timeout 15
Frames
Tree
RobotModel
Status: Ok
Visual Enabled
Collision Enabled
Mass Properties
Update Interval 0
Alpha 1
Description Sou... Topic
Description File
Description Topic /robot_description
Depth 5
History Policy Keep Last
Reliability Po... Reliable
Durability Po... Volatile
TF Prefix
Links
Description Topic
Topic where filepath to urdf is published.
Add
Duplicate
Remove
Rename
Views
Type: Orbit (rviz_del
Zero
Current View Orbit (rviz)
Near Clip ... 0.01
Enable St...
Invert Z A...
Target Fr... <Fixed Frame>
Distance 0.879584
Focal Sha... 0.05
Focal Sha...
Yaw 5.75859
Pitch -1.4148
Focal Point 0; 0; 0
Save
Remove
Rename
Time
ROS Time: 1773126009.18
ROS Elapsed: 1004.24
Wall Time: 1773126009.33
Wall Elapsed: 1004.24
Experimental
Reset
12 fps

Lab 4B

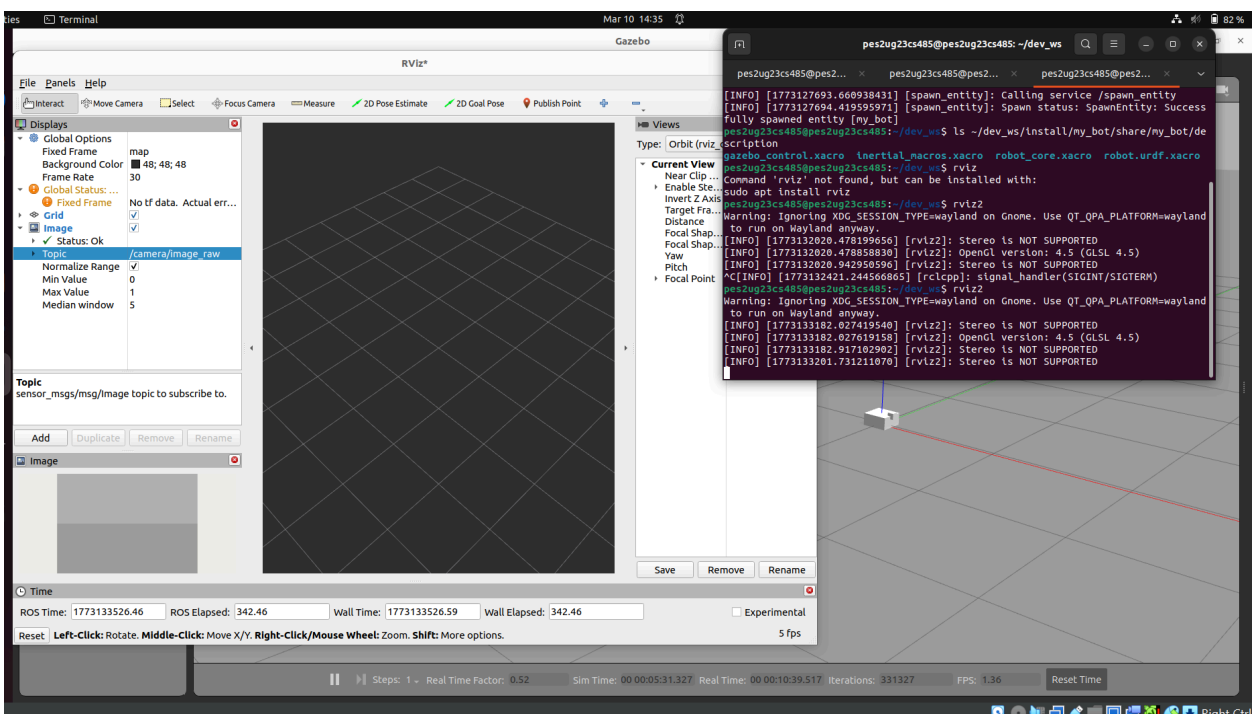

pes2ug23cs485@pes2ug23cs485: ~/dev_ws
pes2ug23cs485:~/dev_ws$ source ~/dev_ws/install/s
pes2ug23cs485:~/dev_ws$ ros2 run gazebo_ros spawn
entity my_bot
[spawn_entity]: Spawn Entity star
[spawn_entity]: Loading entity pu
[spawn_entity]: Waiting for entit
[spawn_entity]: Waiting for servi
[spawn_entity]: Waiting for servi
[spawn_entity]: Calling service /
[spawn_entity]: Spawn status: Spa
[my_bot]
pes2ug23cs485:~/dev_ws$
Gazebo
File Edit Camera View Window Help
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
World Insert Layers
GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
ground_plane
LINKS
link
my_bot
Lights
Property Value
name ground_plane
is_static True
self_coll... False
enable_... False
pose
link
Real Time Factor: 0.61
Sim Time: 00 00:07:42.133
Real Time: 00 00:11:48.003
Iterations: 462133
FPS: 3.20


pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Mar 10 14:08 96 %
pes2ug23cs485@pes2ug23cs485: ~/dev_ws
Moving around:
u i o
j k l
m , .
For Holonomic mode (strafing), hold down the shift key:
U I O
J K L
M < >
t : up (+z)
b : down (-z)
anything else : stop
q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%
CTRL-C to quit
currently: speed 0.50 turn 1.00
Gazebo
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
link
Steps: 1 Real Time Factor: 0.63 Sim Time: 00 00:28:14.762 Real Time: 00 00:52:09.293 Iterations: 1694762 FPS: 4.16 Reset Time
Right Ctrl

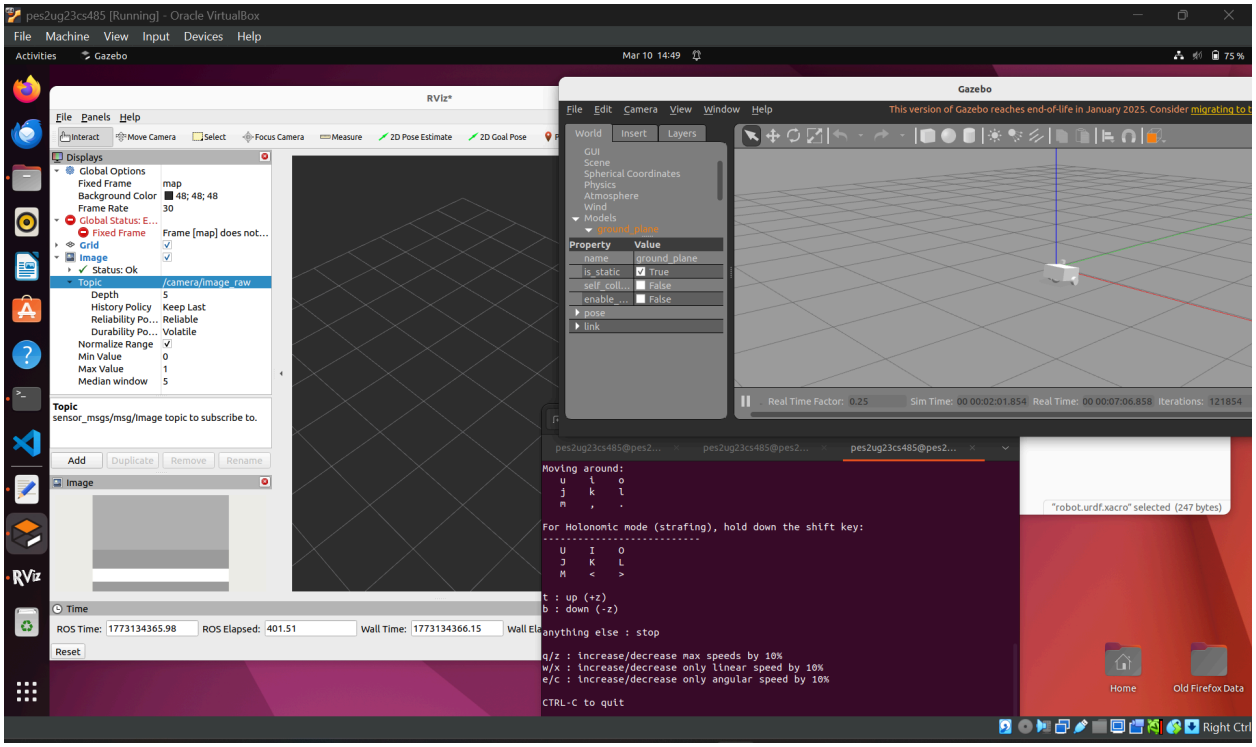

pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Mar 10 14:08 96 %
pes2ug23cs485@pes2ug23cs485: ~/dev_ws
Moving around:
u i o
j k l
m , .
For Holonomic mode (strafing), hold down the shift key:
U I O
J K L
M < >
t : up (+z)
b : down (-z)
anything else : stop
q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%
CTRL-C to quit
currently: speed 0.50 turn 1.00
Gazebo
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
link
Steps: 1 Real Time Factor: 0.73 Sim Time: 00 00:28:26.021 Real Time: 00 00:52:29.416 Iterations: 1706021 FPS: 4.76 Reset Time
Right Ctrl

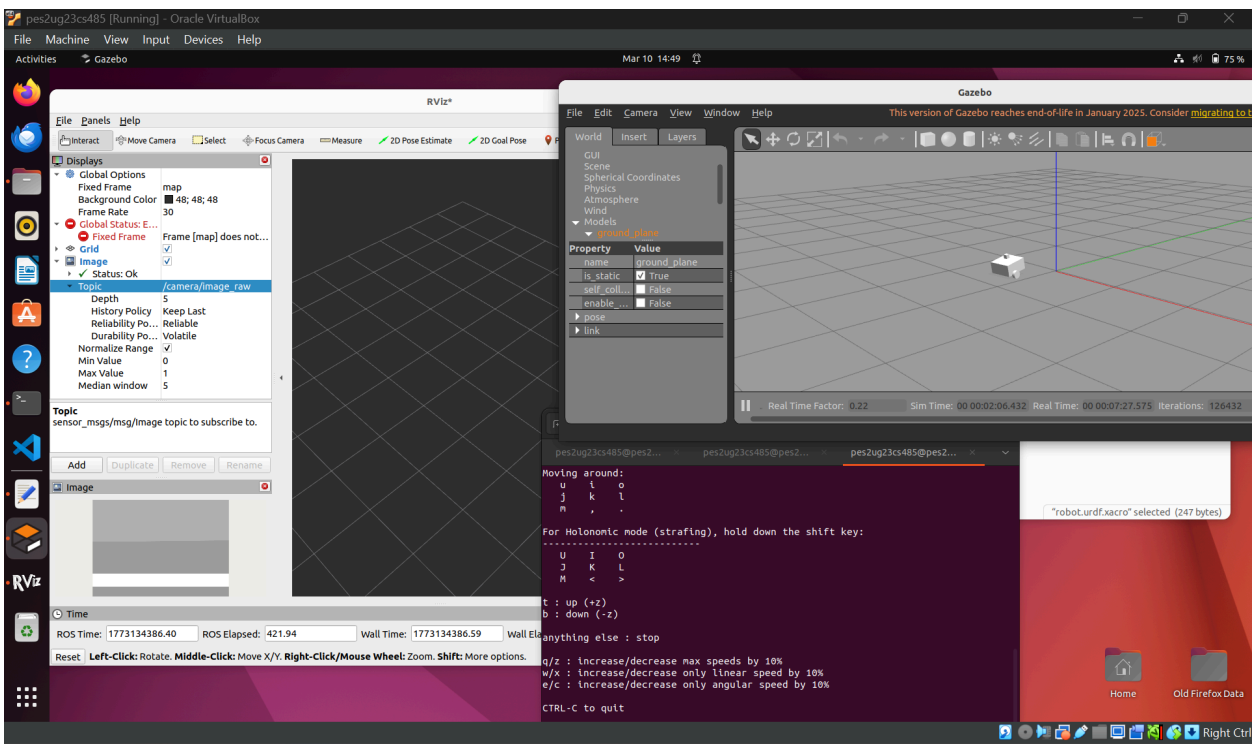

pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities
rviz2
Mar 10 14:11
95 %
pes2ug23cs485@pes2ug23cs485: ~/dev_ws
Gazebo
RViz*
Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
Views
Type: Orbit (rviz_defa
Zero
Current View
Orbit (rviz)
Near Clip ...
0.01
Enable Ste...
Invert Z Axis
Target Fra...
<Fixed Frame>
Distance
10
Focal Shap...
0.05
Focal Shap...
Yaw
5.95858
Pitch
0.225398
Focal Point
0; 0; 0
odom
48; 48; 48
30
OK
Save
Remove
Rename
ROS Elapsed: 91.76
Wall Time: 1773132113.51
Wall Elapsed: 91.76
Experimental
5 fps
Steps: 1
Real Time Factor: 0.38
Sim Time: 00 00:29:51.496
Real Time: 00 00:55:33.974
Iterations: 1791496
FPS: 1.87
Reset Time

Lab 4c


Terminal
Mar 10 14:35
82 %
Gazebo
RViz*
File Panels Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
Displays
Global Options
Fixed Frame
map
Background Color
48; 48; 48
Frame Rate
30
Global Status: ...
Fixed Frame
No tf data. Actual err...
Grid
Image
Status: Ok
Topic
/camera/image_raw
Normalize Range
Min Value
0
Max Value
1
Median window
5
Topic
sensor_msgs/msg/Image topic to subscribe to.
Add
Duplicate
Remove
Rename
Image
Views
Type: Orbit (rviz_
Current View
Near Clip ...
Enable Ste...
Invert Z Axis
Target Fra...
Distance
Focal Shap...
Focal Shap...
Yaw
Pitch
Focal Point
Save
Remove
Rename
Time
ROS Time: 1773133526.46
ROS Elapsed: 342.46
Wall Time: 1773133526.59
Wall Elapsed: 342.46
Experimental
Reset
Left-Click: Rotate. Middle-Click: Move X/Y. Right-Click/Mouse Wheel: Zoom. Shift: More options.
5 fps
pes2ug23cs485@pes2ug23cs485: ~/dev_ws
[INFO] [1773127693.660938431] [spawn_entity]: Calling service /spawn_entity
[INFO] [1773127694.419595971] [spawn_entity]: Spawn status: SpawnEntity: Success fully spawned entity [my_bot]
pes2ug23cs485@pes2ug23cs485:~/dev_ws$ ls ~/dev_ws/install/my_bot/share/my_bot/description
gazebo_control.xacro inertial_macros.xacro robot_core.xacro robot.urdf.xacro
pes2ug23cs485@pes2ug23cs485:~/dev_ws$ rviz
Command 'rviz' not found, but can be installed with:
sudo apt install rviz
pes2ug23cs485@pes2ug23cs485:~/dev_ws$ rviz2
Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.
[INFO] [1773132020.478199656] [rviz2]: Stereo is NOT SUPPORTED
[INFO] [1773132020.478858830] [rviz2]: OpenGl version: 4.5 (GLSL 4.5)
[INFO] [1773132020.942950596] [rviz2]: Stereo is NOT SUPPORTED
^C[INFO] [1773132421.244566865] [rclcpp]: signal_handler(SIGINT/SIGTERM)
pes2ug23cs485@pes2ug23cs485:~/dev_ws$ rviz2
Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.
[INFO] [1773133182.027419540] [rviz2]: Stereo is NOT SUPPORTED
[INFO] [1773133182.027619158] [rviz2]: OpenGl version: 4.5 (GLSL 4.5)
[INFO] [1773133182.917102902] [rviz2]: Stereo is NOT SUPPORTED
[INFO] [1773133201.731211070] [rviz2]: Stereo is NOT SUPPORTED
Steps: 1
Real Time Factor: 0.52
Sim Time: 00 00:05:31.327
Real Time: 00 00:10:39.517
Iterations: 331327
FPS: 1.36
Reset Time

Lab 4d


pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Gazebo Mar 10 14:49 75 %
RViz*
File Panels Help
Interact Move Camera Select Focus Camera Measure 2D Pose Estimate 2D Goal Pose
Displays
Global Options
Fixed Frame map
Background Color 48; 48; 48
Frame Rate 30
Global Status: E...
Fixed Frame Frame [map] does not...
Grid
Image
Status: Ok
Topic /camera/image_raw
Depth 5
History Policy Keep Last
Reliability Po... Reliable
Durability Po... Volatile
Normalize Range
Min Value 0
Max Value 1
Median window 5
Topic
sensor_msgs/msg/Image topic to subscribe to.
Add Duplicate Remove Rename
Image
Time
ROS Time: 1773134365.98 ROS Elapsed: 401.51 Wall Time: 1773134366.15
Reset
Gazebo
File Edit Camera View Window Help
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to t
World Insert Layers
GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
ground_plane
Property Value
name ground_plane
is_static True
self_coll... False
enable_... False
pose
link
Real Time Factor: 0.25 Sim Time: 00 00:02:01.854 Real Time: 00 00:07:06.858 Iterations: 121854
pes2ug23cs485@pes2...
Moving around:
u i o
j k l
m , .
For Holonomic mode (strafing), hold down the shift key:
U I O
J K L
M < >
t : up (+z)
b : down (-z)
anything else : stop
q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%
CTRL-C to quit
"robot.urdf.xacro" selected (247 bytes)
Home
Old Firefox Data
Right Ctrl


pes2ug23cs485 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Gazebo Mar 10 14:49 75 %
RViz*
File Panels Help
Interact Move Camera Select Focus Camera Measure 2D Pose Estimate 2D Goal Pose
Displays
Global Options
Fixed Frame map
Background Color 48; 48; 48
Frame Rate 30
Global Status: E...
Fixed Frame Frame [map] does not...
Grid
Image
Status: Ok
Topic /camera/image_raw
Depth 5
History Policy Keep Last
Reliability Po... Reliable
Durability Po... Volatile
Normalize Range
Min Value 0
Max Value 1
Median window 5
Topic
sensor_msgs/msg/Image topic to subscribe to.
Add Duplicate Remove Rename
Image
Time
ROS Time: 1773134386.40 ROS Elapsed: 421.94 Wall Time: 1773134386.59
Reset Left-Click: Rotate. Middle-Click: Move X/Y. Right-Click/Mouse Wheel: Zoom. Shift: More options.
Gazebo
File Edit Camera View Window Help
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to t
World Insert Layers
GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
ground_plane
Property Value
name ground_plane
is_static True
self_coll... False
enable_... False
pose
link
Real Time Factor: 0.22 Sim Time: 00 00:02:06.432 Real Time: 00 00:07:27.575 Iterations: 126432
pes2ug23cs485@pes2...
Moving around:
u i o
j k l
m , .
For Holonomic mode (strafing), hold down the shift key:
U I O
J K L
M < >
t : up (+z)
b : down (-z)
anything else : stop
q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%
CTRL-C to quit
"robot.urdf.xacro" selected (247 bytes)
Home
Old Firefox Data
Right Ctrl